

Motion Planning and Incremental Nonlinear Dynamic Inversion for Aggressive Quadrotor Trajectory Tracking on the Crazyflie 2.x Platform Family

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Abstract

This report describes the design, implementation, and experimental evaluation of a motion planning and control pipeline for quadrotors: minimum-snap motion planning with automatic time allocation, a differential-flatness-based geometric SE(3) baseline controller, and an Incremental Nonlinear Dynamic Inversion (INDI) attitude-and-position controller. All three components were deployed and compared across three quadrotor platforms: a standard Crazyflie 2.1, a thrust-upgraded Crazyflie 2.1, and a Crazyflie 2.1 Brushless (CF21BL). Each platform was commissioned end-to-end (motor identification via RPM, filter tuning, cascaded gain tuning) and evaluated on a shared figure-8 tracking benchmark, plus oval and circle trajectories swept across a range of planner aggressiveness. The main result is a controller-and-platform comparison showing that full INDI reduces XY tracking RMSE by 37–60% relative to the geometric SE(3) baseline. The brushless platform’s higher motor bandwidth ($\tau \approx 44$ ms vs. 71 ms for the thrust-upgraded platform) allows more aggressive attitude gains and gives the best overall tracking performance (2.45 cm mean RMSE, $n = 13$), but also brings a low-frequency, gain-driven attitude resonance at ≈ 7.2 Hz, close to the loop’s own natural frequency $\omega_n = \sqrt{K_R}$, that the thrust-upgraded platform’s lower bandwidth avoids. A targeted band-reject (notch) filter, implemented and flight-tested as a fix, did not resolve it – consistent with a gain-driven resonance rather than noise a filter can remove. The remaining sections cover related work, methods, experimental setup, results, a discussion of the cross-platform differences, and the project’s open limitations.

1 Introduction

Tracking aggressive trajectories accurately relies on two complementary techniques. The first is **differential-flatness-based motion planning**, which produces dynamically feasible, minimum-effort trajectories together with closed-form full-state references derived directly from the position trajectory and its derivatives. The second is **Incremental Nonlinear Dynamic Inversion (INDI)**, a control technique that replaces an explicit dynamics model with a real-time measurement of the vehicle’s actual angular acceleration, so the attitude loop stays robust to unmodeled effects and platform-specific uncertainty.

This project implements both. A minimum-snap planning pipeline with automatic time allocation (Richter et al. [1]) generates the trajectories, and full INDI attitude-and-position control (Tal & Karaman [2]), including the RPM-feedback-based torque estimation it requires, tracks them on real Crazyflie hardware. We commissioned the full stack end to end (motor identification via RPM, filter tuning, cascaded gain tuning) on three physical platforms: a standard, a thrust-upgraded, and a brushless CF21BL Crazyflie 2.x. This lets us see how platform-specific actuator dynamics limit the achievable control bandwidth and tracking accuracy. Each platform is bench-

marked against a common geometric SE(3) baseline and pushed across a planner-aggressiveness sweep (k_t) to find its practical ceiling. Two originally planned phases were dropped once a data-confirmed attitude resonance on the highest-bandwidth platform became the project’s main open question (Section 6.2): commissioning a third platform, and a dedicated accel-pinned aerobatics phase. Part of that aerobatics work, a fully inverted loop, was still completed as bonus work once time allowed (Section 7), using the same planning modes as every other experiment (Section 3.1) with an added acceleration constraint at selected waypoints – attitude is still recovered purely from flatness, never authored directly.

2 Related Work

A quadrotor’s translational dynamics are differentially flat [3]: the full 6-DOF state and the control inputs can be written as algebraic functions of the position trajectory $\mathbf{p}(t)$, yaw $\psi(t)$, and a finite number of their derivatives (up to snap for the full state, one derivative further for angular-acceleration feedforward). A smooth polynomial in position/yaw space is therefore enough to derive a complete, dynamically consistent reference without ever planning attitude directly. Our baseline controller follows the geometric SE(3) approach of Lee et al. [4]: it tracks the flatness-derived desired attitude R_d by computing attitude and rate errors on SO(3), avoiding Euler-angle singularities, and commands torque model-based from these errors plus a gyroscopic decoupling term. Trajectories are generated with Mellinger and Kumar’s [3] minimum-snap QP formulation, extended by Richter et al. [1] to automate per-segment timing from a single aggressiveness scalar k_t . The INDI attitude law (Tal & Karaman [2]) replaces the geometric law’s model-based feedback with an incremental update: the next torque command is the currently applied torque plus a correction proportional to the gap between the desired and the measured angular acceleration. Because any unmodeled effect shows up directly in that measurement, INDI stays robust to model mismatch without needing the mismatch to be characterized in advance. This property turns out to matter a great deal for the main practical challenge of this project (Section 6.2).

3 Methods

3.1 Motion planning

All trajectories in this project are generated by one of two automatic-time-allocation planning modes. Both are built on the same minimum-snap QP machinery (`src/planning/richter.rs`) and both are driven by a single scalar aggressiveness parameter k_t rather than manually chosen segment durations:

- **Mode 1** (`RichterTrajectory::plan()`) uses Richter et al.’s [1] time-allocation formula,

$$T_i = \max\left(\frac{\text{dist}_i}{v_{\text{avg}}(k_t)}, T_{\min}\right), \quad v_{\text{avg}} = 0.5 + 1.5\sqrt{k_t}, \quad T_{\min} = 0.15 \text{ s}$$

below which the QP’s $1/T^7$ segment weighting becomes numerically unstable. We use zero redistribution iterations by default, favoring robustness and predictability over the marginal timing refinement the full paper formulation offers.

- **Mode 3** (`RichterTrajectory::plan_paper()`) uses the same time-allocation formula but adds the full eight gradient-descent redistribution iterations specified in Richter et al. [1], which push segment time toward the highest-snap portions of the trajectory for more balanced actuator demand across the maneuver.

Both modes share the same differential-flatness recovery, following the Faessler et al. [5] formulation (implemented in `src/planning/flatness.rs`). Given the position polynomial’s derivatives

at any query time t (velocity, acceleration, jerk, snap) plus yaw and its derivatives, the full rigid-body reference is recovered in closed form:

$$\mathbf{a}_g = \ddot{\mathbf{p}} + g\hat{\mathbf{z}} \quad (1)$$

$$\mathbf{x}_b = \text{normalize}(\mathbf{y}_c \times \mathbf{a}_g), \quad \mathbf{y}_b = \text{normalize}(\mathbf{a}_g \times \mathbf{x}_b), \quad \mathbf{z}_b = \mathbf{x}_b \times \mathbf{y}_b \quad (2)$$

$$c = \mathbf{z}_b \cdot \mathbf{a}_g, \quad T = m c \quad (3)$$

$$\boldsymbol{\omega} = f(\mathbf{j}, \mathbf{x}_b, \mathbf{y}_b, c) \quad (\text{from jerk } \mathbf{j} = \ddot{\mathbf{p}}) \quad (4)$$

$$\dot{\boldsymbol{\omega}} = f(\mathbf{s}, \boldsymbol{\omega}, \mathbf{x}_b, \mathbf{y}_b, c) \quad (\text{from snap } \mathbf{s} = \mathbf{p}^{(4)}) \quad (5)$$

$$\boldsymbol{\tau} = J\dot{\boldsymbol{\omega}} - (J\boldsymbol{\omega}) \times \boldsymbol{\omega} \quad (6)$$

Here $\mathbf{x}_c = [\cos \psi, \sin \psi, 0]$ and $\mathbf{y}_c = [-\sin \psi, \cos \psi, 0]$ define the yaw-heading intermediate frame used to disambiguate the body- x /body- y axes from the thrust-direction-only $\mathbf{z}_b$. This closed-form map lets the entire trajectory library used in this project (figure-8, circle, oval, corner, helix, tilted oval) be authored purely as position/yaw waypoints. Attitude, angular rate, angular acceleration, and thrust are all derived from that, never planned separately.

3.2 Geometric SE(3) baseline control

The geometric controller [4] is the same control law used across all three platforms and is INDI's structural baseline throughout this report. The formal, validated geometric-vs-INDI comparison in Section 5.2 (Table 1) was carried out on the standard platform only; the thrust-upgraded and brushless platforms are compared to each other and to that standard-platform reference through their own INDI results, not through their own separate geometric campaigns. Attitude error is computed on SO(3):

$$\mathbf{e}_R = \frac{1}{2} \text{vee} \left(R_d^\top R - R^\top R_d \right)$$

and torque is commanded directly from a model-based law:

$$\boldsymbol{\tau} = -K_R \mathbf{e}_R - K_W \mathbf{e}_\omega + \boldsymbol{\omega} \times J\boldsymbol{\omega}$$

with K_R and K_W tuned per platform (the position loop uses K_P/K_V on position/velocity error with the same cascade structure). Because this law is purely model-based, its accuracy depends on how closely J (inertia) and the thrust/torque allocation model match the real airframe.

3.3 INDI: attitude loop

Full INDI has two loops, attitude and position; this subsection covers the attitude loop (the position loop follows in Section 3.4). The attitude-loop law, following Tal & Karaman [2] directly, replaces the geometric law's model-based torque computation with four coupled equations.

3.3.1 (1) Feedforward

The desired angular acceleration $\boldsymbol{\alpha}_{\text{des}}$ is computed from the trajectory's snap via differential flatness (Tal & Karaman [2], Eq. 15). This is the only place the desired trajectory's higher derivatives feed the attitude loop; everything else below is feedback.

3.3.2 (2) Reference blend

$$\boldsymbol{\alpha}_{\text{ref}} = \boldsymbol{\alpha}_{\text{des}} - K_R \mathbf{e}_R - K_W \mathbf{e}_\omega \quad (\text{Eq. 28})$$

This has the same form as the geometric law's gains, but $\boldsymbol{\alpha}_{\text{ref}}$ is a desired acceleration, not a torque. K_R has units $1/\text{s}^2$ (closed-loop bandwidth $\omega_n = \sqrt{K_R}$) and K_W has units $1/\text{s}$ (damping ratio $\zeta = K_W/(2\omega_n)$).

3.3.3 (3) Measured angular acceleration

No sensor measures angular acceleration directly, so α_{meas} is derived from the gyroscope through a two-stage chain, matching the order specified by Tal & Karaman [2] Sec. III-D and the stock Crazyflie firmware: the raw body rate is first low-pass filtered, then differentiated.

$$\omega_{\text{filt}}[k] = \text{BW}(\omega_{\text{raw}}[k]), \quad \alpha_{\text{meas}}[k] = \frac{\omega_{\text{filt}}[k] - \omega_{\text{filt}}[k-1]}{\Delta t}$$

The same Butterworth filter is applied identically to α_{ref} , which keeps the reference and measured legs phase-matched before their difference is taken in the increment law below.

3.3.4 (4) Incremental torque law

$$\delta \tau = J(\alpha_{\text{ref}} - \alpha_{\text{meas}}) \quad (7)$$

$$\tau_{\text{new}} = \tau_{\text{current}} + \delta \tau \quad (\text{Eq. 31}) \quad (8)$$

This is the core INDI mechanism: a small correction to whatever torque is currently applied, using measured, not modeled, angular acceleration. τ_{current} is estimated from the physical system, per-motor RPM converted to thrust via a bench-identified motor constant k_T and then mapped to body torque via the known arm geometry, rather than from the previous commanded torque, which would not account for real motor response lag. This is the key structural difference from the geometric law: INDI does not need an accurate model of thrust-to-RPM response, aerodynamic drag, or other unmodeled disturbances, because any such effect already shows up in α_{meas} and is rejected by the increment on the next cycle.

3.4 INDI: position loop

The same incremental principle from the attitude loop above also drives the position loop (`ctrl_mode` bit 0, active alongside the attitude bit on all three locked configurations, Appendix B): instead of trusting the mass/thrust model for the feedforward acceleration, it measures the mismatch between that model’s predicted and the IMU’s actual net acceleration, and adds the mismatch directly to the desired acceleration, $\mathbf{f}_d = \mathbf{a}_d + K_P \mathbf{e}_p + K_V \mathbf{e}_v + \mathbf{a}_{\text{indi}}$ where $\mathbf{a}_{\text{indi}} = \mathbf{a}_{\text{meas}} - \mathbf{a}_{\text{model}}$. Same robustness argument as the attitude loop: unmodeled thrust/mass error shows up directly in that mismatch and is corrected next cycle rather than characterized in advance. Full equations in Appendix A.

3.5 An added filter stage: an attempted fix for a diagnosed narrowband resonance

Brushless-platform commissioning identified a narrowband attitude oscillation at ≈ 7.22 Hz (Section 6.2 gives the full root-cause analysis), which the existing Stage-1 Butterworth filter does not attenuate. As a candidate fix we added an optional, runtime-tunable second-stage band-reject (notch) filter, applied identically to α_{meas} , α_{ref} , and τ_{current} so no term of the INDI sum ends up phase-mismatched (full filter design in Appendix A). It is disabled by default (`notch_en=0`, byte-identical to prior behavior when off) and was implemented, build-verified, and flight-tested across a range of bandwidths – it did not resolve the resonance (Section 7).

4 Experimental Setup

We commissioned three platforms: a **standard** Crazyflie 2.1 (brushed 0716 motors, 36.4 g, $J_{xx} = J_{yy} = 1.657 \times 10^{-5} \text{ kg m}^2$, used as the reference platform), a **thrust-upgraded** Crazyflie 2.1

(brushed upgrade kit, 38.6 g, similar inertia, higher peak thrust per motor at 0.18 N vs. 0.13 N, but a heavier rotor), and a **brushless** CF21BL (41.0 g, $\text{diag}(2.40, 2.40, 3.24) \times 10^{-5} \text{ kg m}^2$ [6], the highest motor bandwidth of the three, with DShot bidirectional telemetry). Full physical parameters are listed in Appendix B (Table 2). Every platform was bench-identified for its per-motor thrust constant k_T (thrust stand, RPM sweep, $k_T = \text{thrust}/\text{RPM}^2$) before any INDI flight.

Trajectories are generated offline in Rust (`export_poly4d`), converted to an 8-coefficient Poly4D representation, and uploaded via the CrazySwarm2 (CS2) ROS 2 framework. Position is estimated from OptiTrack and fed to the onboard EKF. The onboard high-level commander evaluates the uploaded polynomial at 1 kHz and feeds the geometric or INDI controller, running at 500 Hz, through the standard Crazyflie firmware loop. This gives globally drift-free pose for repeatable long-duration, high-speed flights.

The primary metric throughout is **phase-aligned XY position RMSE**: the flown path is time-shifted to maximize cross-correlation with the planned path before computing RMSE, which corrects for any constant commanded-vs-flown lag. Secondary metrics are roll/pitch tracking error and, for stress-testing, per-axis gyro standard deviation as a proxy for attitude oscillation severity. We classify a flight as a **crash** (used throughout Section 5.3 and Appendix D) when logged altitude dives well below the commanded trajectory, which is physically impossible under controlled flight, and this is corroborated by a simultaneous jump in gyro standard deviation. Both signals are unambiguous in the logged telemetry and were cross-checked against each flight’s own recording. All exported trajectories were validated against a physically confirmed usable flight volume ($x: \pm 1.2 \text{ m}$, $y: \pm 2.5 \text{ m}$, $z: 0.5\text{--}1.8 \text{ m}$) by densely sampling every trajectory across the full k_t range flown and checking worst-case bounding boxes before any flight.

The core benchmark for every cross-platform, cross-controller comparison is a **figure-8** at $k_t = 0.05$ (lap $\approx 6.9 \text{ s}$). Extended aggressiveness sweeps additionally use **oval** and **circle** shapes at increasing k_t to find each platform’s practical ceiling.

5 Results

5.1 Per-platform tuning summary

Full locked-configuration tables (all gains, filters, and bench-identified thrust constants) are in Appendix B (Tables 3–5).

Standard CF2.1. A filter cutoff sweep found $\text{fc_bw} = 70 \text{ Hz}$ optimal (3.1 cm RMSE). The final configuration is $K_R = 1050$, $K_W = 87$ ($\omega_n \approx 32.4 \text{ rad/s}$), $K_{P,xy} = 40$, $K_{V,xy} = 3.3$. Over 10 flights, INDI averaged **3.87 cm** RMSE against 6.14 cm for the geometric baseline, a 37% reduction (Figure 1, Appendix C).

Thrust-upgraded CF2.1. The attitude gain ladder found a hard ceiling at $K_R = 603$ ($\omega_n \approx 24.6 \text{ rad/s}$), well below the standard platform’s 1050, despite the upgrade kit’s higher peak thrust (0.18 N vs. 0.13 N per motor). The final configuration is $K_R = 603$, $K_W = 90$ ($\zeta \approx 1.83$, $\tau_{\text{slow}} = 137 \text{ ms}$), $K_{P,xy} = 40$, $K_{V,xy} = 8$. An 11-flight validation gave **3.72 cm** mean RMSE (std 0.21), close to the standard platform despite the lower achievable K_R .

Brushless CF21BL. This platform saw the most extensive tuning campaign of the three. A coupled gain ladder (K_R , K_W , K_P , K_V scaled together) improved RMSE monotonically up to a hover-noise ceiling at $K_R \approx 2400\text{--}2600$. Decoupled sweeps then showed that pulling K_W and K_V down independently of that ratio improved both RMSE and attitude error at the same time, a real decoupling win rather than a tradeoff, because K_W injects raw gyro noise directly and linearly through $K_W \mathbf{e}_\omega$. The final configuration is $K_R = 2400$, $K_W = 170$ ($\zeta \approx 1.74$),

Configuration	n	Mean XY RMSE	Std	vs. Standard Geometric
Standard Geometric (baseline)	10	6.14 cm	0.52	—
Standard INDI	10	3.87 cm	0.24	−37%
Upgraded INDI	11	3.72 cm	0.21	−39%
Brushless INDI (fc_bw=60)	12	2.48 cm	0.13	−60%
Brushless INDI (fc_bw=70)	13	2.45 cm	0.12	−60%

Table 1: Five-way headline comparison, figure-8 trajectory, $k_t = 0.05$.

$K_{P,xy} = 64$, $K_{V,xy} = 5$. A 12-flight validation gave **2.48 cm** mean RMSE (std 0.14), the best of any platform in this project.

5.2 Headline cross-platform/cross-controller comparison

The fc_bw = 60 vs. 70 Hz brushless columns are otherwise identical configurations, so they isolate the filter cutoff’s effect: the two means (2.48 ± 0.14 vs. 2.45 ± 0.12 cm) fall well within one combined standard deviation of each other across the 20–100 Hz range tested (we did not run a formal significance test). This is consistent with this platform’s tracking ceiling being set by attitude gain, and the resonance it induces (Section 6.2), rather than filter tuning. A bar-chart visualization (Figure 2) and the full per-flight breakdown of the best brushless flight (flown path, 3D attitude, INDI torque, and RPM balance) are reproduced at full size in Appendix C (Figures 3–6).

5.3 Extended aggressiveness sweeps (upgraded vs. brushless)

Full per- k_t data (XY/Z RMSE, roll/pitch/yaw error, max speed) for both sweeps is in Appendix D (Tables 6–7). Unlike the headline comparison above ($n = 10$ –13 flights), each k_t point here is only 1–3 flights – enough for the qualitative clean/crash boundary, not fine-grained statistics.

Oval. Both platforms flew cleanly up to $k_t \leq 0.3$ –0.4. The upgraded platform crashed consistently at $k_t = 0.5$ –0.6 (5/5 attempts), while brushless stayed clean through $k_t = 0.5$ and crashed only at $k_t = 0.7$. Brushless’s working ceiling is roughly $k_t \approx 0.5$ against upgraded’s $k_t \approx 0.3$ –0.4. This tracks the bandwidth gap between them ($\omega_n \approx 7.8$ Hz for brushless vs. ≈ 3.9 Hz for upgraded), which gives brushless more control authority to correct through sharp direction changes, at the cost of sitting closer to its own resonant regime.

Circle. This shape was newly swept on both platforms in this project; previously only $k_t = 0.05$ had been flown. Both platforms show the same pattern: a clean-flight region followed by a marginal zone where crash and clean outcomes mix at the same k_t rather than switching at a sharp threshold. Upgraded stays clean through $k_t = 0.8$ (an isolated crash at $k_t = 0.9$ notwithstanding), while brushless holds clean through $k_t = 0.7$ before crashing at 0.8–0.9. Cross-platform RMSE at matched k_t is tied at $k_t = 0.05$ (1.7 vs. 1.9 cm) but favors brushless at $k_t = 0.3$ (1.4 vs. 2.2 cm, about 36% lower), so which platform tracks better depends on k_t rather than being a fixed ranking.

Tracking-overview plots for both representative points (oval $k_t = 0.4$, circle $k_t = 0.7$) are reproduced at full size in Appendix C (Figures 7–8).

6 Discussion

6.1 Why the platforms rank the way they do

Considering position RMSE together with attitude tracking error gives a clearer ranking than RMSE alone. In the best-flight breakdown (Appendix C, Figure 3), roll error runs 7.2° (standard) $\rightarrow 5.3^\circ$ (upgraded) $\rightarrow 4.9^\circ$ (brushless), and pitch error $5.1^\circ \rightarrow 3.4^\circ \rightarrow 3.2^\circ$ in the same order, with position RMSE following the same order in that flight (3.6, 3.2, 2.2 cm). So on both metrics, brushless tracks best, thrust-upgraded second, standard third, even though the aggregate n -flight RMSE in Table 1 shows standard and upgraded close enough (3.87 vs. 3.72 cm) to fall within one combined standard deviation of each other.

Brushless’s clear lead is well explained by achievable INDI attitude bandwidth, set by motor response speed rather than peak thrust: its actuator time constant ($\tau \approx 44$ ms) is faster than the thrust-upgraded platform’s ($\tau \approx 71$ ms), and a faster actuator supports a higher stable K_R (2400 vs. 603) and therefore higher closed-loop bandwidth – a hardware-set advantage, not a tuning outcome. Brushless leads even with the ≈ 7.22 Hz resonance below still unresolved; fixing it should widen this lead, not shrink it. The smaller gap between standard and upgraded does not fit as cleanly: standard reaches a higher stable K_R (1050 vs. 603) yet tracks slightly worse on every metric above. No confirmed explanation for this reversal is available from the present data; we report it as an open observation rather than propose an explanation the evidence does not support.

6.2 The brushless attitude resonance: what we observed and our working hypothesis

Two things are directly observed, not inferred. First, the brushless platform’s raw angular-acceleration signal (α_{raw}) shows a narrowband spectral peak at ≈ 7.22 Hz, measured by 500 Hz SD-card logging. The existing Stage-1 Butterworth filter does essentially nothing to it ($|\text{filtered}|/|\text{raw}| = 1.000$), since its cutoff (60–70 Hz) sits well above this band by design. Second, commanded torque on brushless is markedly larger than on the other two platforms across all three axes in the same flight comparison (Appendix C, Figure 5): τ_z RMS is 2.17 vs. 0.19 (standard) and 0.14 (upgraded) N·m, roughly an order of magnitude higher. Beyond these two observations, what follows is our working hypothesis, not a confirmed cause. ≈ 7.22 Hz is comparatively low for mechanical/aerodynamic noise, which typically shows up higher, tied to prop/motor RPM; our supervisor’s assessment was that a peak this low points toward a gain/bandwidth-driven mode rather than noise, matching our leading candidate: a feedback resonance, $\omega_n = \sqrt{K_R}/2\pi = 7.80$ Hz at $K_R = 2400$, close to the measured 7.22 Hz peak (within 8%), consistent with the closed-loop bandwidth sitting near its own resonant frequency and sustaining a self-oscillation there (Barkhausen/Nyquist condition). Converting the bench-measured actuator lag (31.6–33.6 ms) to phase at 7.22 Hz gives $\approx 82^\circ$, about four times the $\approx 20^\circ$ from EKF attitude-estimate lag, so actuator lag is the larger of the two measured contributors if this picture is right. Lowering the filter cutoff into the resonant band made tracking measurably worse, consistent with a low-pass trading off phase margin across its whole passband to reach one narrow region – none of this proves the mechanism on its own. The full 500 Hz diagnostic is in Appendix C (Figures 9–10), showing the same peak at hover and under real maneuvering load. Two follow-up flight tests (Section 7) add evidence without proving it: the notch filter made tracking worse, fitting a gain-driven resonance better than noise, and lowering K_R also failed, for an unrelated reason.

7 Limitations & Future Work

Brushless attitude resonance (flight-tested, unresolved). The notch filter was implemented, build-verified, and flight-tested in hover only, not on any trajectory, across 0.25–20 Hz bandwidth centered on the measured ≈ 7 Hz peak. Several configurations destabilized the loop

outright; among those that did not, tracking was still worse than unfiltered – the filter measurably attenuated the target band in the signal itself, but that did not translate into better closed-loop behavior. $\omega_n = \sqrt{k_r} \approx 7.8 \text{ Hz}$ matches the measured peak closely: a low-frequency, gain-driven resonance, not high-frequency noise a filter can remove, so attenuating it in the error signal removes stabilizing feedback instead. Lowering k_r also failed, triggering divergence from a violated position/attitude cascade timescale separation. The most direct remaining path – reducing the phase lag sustaining the oscillation (actuator lag is the larger contributor, Section 6.2) or restructuring the cascade for a lower k_r – is beyond scope here.

Inverted-loop aerobatic maneuver (bonus, beyond core scope). Using the same planning modes as every other experiment (Section 3.1) with an added acceleration constraint at selected waypoints (attitude still purely from flatness, never authored directly), we commissioned a fully inverted vertical loop on the brushless platform. Pre-flip works as expected: a repeatable near-full inversion (163–178° at the apex, 3/3 flights) followed by genuine recovery, arresting the fall and climbing back (Appendix C, Fig. 11). Post-flip is still unresolved: the drone tumbles again in the trajectory’s auto-generated exit ramp, and recovering from it has not yet worked despite several fixes tried and ruled out. **A third platform and a dedicated aggressive-maneuver phase** were both descopeed once the resonance investigation became dominant; the inverted-loop work above is a partial, unplanned recovery of the latter.

8 Conclusion

This project implemented minimum-snap motion planning with automatic time allocation, a differential-flatness-derived geometric baseline, and INDI attitude-and-position control, then commissioned and evaluated all three on three physically distinct quadrotor platforms. The central result is a 37–60% reduction in figure-8 tracking RMSE from INDI over the geometric baseline, governed mostly by each platform’s motor bandwidth, not tuning effort. The trajectory library was pushed beyond the core benchmark with oval and circle sweeps, mapping out real, platform-specific speed ceilings. The most significant open finding, a narrowband $\approx 7.22 \text{ Hz}$ attitude resonance on the highest-bandwidth platform, is documented with the observations behind it, a working hypothesis for its mechanism, and a notch-filter fix that was flight-tested but did not resolve it, pointing toward the loop’s own gain-driven resonance, not noise, as the likely remaining lever. The limitations reported here, together with an unplanned excursion into fully inverted aerobatic flight, are well-characterized open problems, not unexplained failures.

References

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A Full INDI and Notch-Filter Equations

This appendix collects the complete equation set summarized in the Methods section, including implementation-level detail (exact filter coefficients, the motor allocation matrix) omitted from the main text for readability.

A.1 Snap-based angular acceleration feedforward

The desired angular acceleration $\boldsymbol{\alpha}_{\text{des}}$ (Tal & Karaman [2], Eq. 15) is computed in closed form from the trajectory's jerk $\mathbf{j} = \ddot{\mathbf{p}}$ and snap $\mathbf{s} = \mathbf{p}^{(4)}$, for constant yaw ($\dot{\psi} = 0$):

$$\dot{c} = \mathbf{b}_3 \cdot \mathbf{j} \quad (9)$$

$$\alpha_x = \frac{2(\mathbf{y}_b \cdot \mathbf{j})(\mathbf{b}_3 \cdot \mathbf{j})}{c^2} - \frac{\mathbf{y}_b \cdot \mathbf{s}}{c} \quad (10)$$

$$\alpha_y = \frac{\mathbf{x}_b \cdot \mathbf{s}}{c} - \frac{2(\mathbf{x}_b \cdot \mathbf{j})(\mathbf{b}_3 \cdot \mathbf{j})}{c^2} \quad (11)$$

$$\alpha_z = 0 \quad (12)$$

where $c = \|\mathbf{a} + g\hat{\mathbf{z}}\|$ and $\mathbf{x}_b, \mathbf{y}_b, \mathbf{b}_3$ are the desired body axes from the flatness recovery (Section 3.1). This is a pure feedforward term computed once per trajectory sample; no filtering is applied to $\boldsymbol{\alpha}_{\text{des}}$ itself, since it is a planned reference signal, not a noisy measurement.

A.2 Reference blend and rate error

$$\mathbf{e}_\omega = \boldsymbol{\omega} - \boldsymbol{\omega}_d \quad (13)$$

$$\boldsymbol{\alpha}_{\text{ref}} = \boldsymbol{\alpha}_{\text{des}} - K_R \mathbf{e}_R - K_W \mathbf{e}_\omega \quad (14)$$

where $\boldsymbol{\omega}_d$ (desired body angular rate) is itself a differential-flatness feedforward term derived from jerk (Tal & Karaman [2], Eq. 14), and $\mathbf{e}_R$ is the SO(3) attitude error defined in Section 3.2.

A.3 Measured angular-acceleration filter chain

Second-order Butterworth (bilinear-transform biquad, cutoff f_c , sample period Δt), applied to the raw body rate $\boldsymbol{\omega}_{\text{raw}}$:

$$y[k] = b(x[k] + 2x[k-1] + x[k-2]) - a_1y[k-1] - a_2y[k-2]$$

with the standard bilinear-transform coefficients b, a_1, a_2 derived from f_c and Δt .

Finite difference, applied to the filtered rate:

$$\boldsymbol{\alpha}_{\text{meas}}[k] = \frac{\boldsymbol{\omega}_{\text{filt}}[k] - \boldsymbol{\omega}_{\text{filt}}[k-1]}{\Delta t}$$

The identical Butterworth filter (same cutoff, independent instances) is applied to $\boldsymbol{\alpha}_{\text{ref}}$ to keep the reference and measured legs of the increment phase-matched before their difference is taken in Eq. (19) below.

A.4 Optional stage-2 notch (band-reject) filter

The added filter of Section 3.5 is a standard RBJ Audio-EQ-Cookbook band-reject biquad, Direct Form I, parameterized by center frequency f_0 and bandwidth bw (both runtime-tunable):

$$\omega_0 = 2\pi f_0 \Delta t, \quad Q = \frac{f_0}{\text{bw}} \quad (15)$$

$$\alpha = \frac{\sin \omega_0}{2Q}, \quad a_0 = 1 + \alpha \quad (16)$$

$$b_0 = b_2 = \frac{1}{a_0}, \quad b_1 = a_1 = \frac{-2 \cos \omega_0}{a_0}, \quad a_2 = \frac{1 - \alpha}{a_0} \quad (17)$$

$$y[k] = b_0 x[k] + b_1 x[k-1] + b_2 x[k-2] - a_1 y[k-1] - a_2 y[k-2] \quad (18)$$

With default $f_0 = 7.2 \text{ Hz}$, $\text{bw} = 5.0 \text{ Hz}$ ($Q \approx 1.44$), the -3 dB attenuated band is approximately $f_0 \pm \text{bw}/2 \approx 4.7\text{--}9.7 \text{ Hz}$, matching the diagnosed shake band. When enabled, this same filter (independent instances, identical coefficients) is applied to $\boldsymbol{\alpha}_{\text{meas}}$, $\boldsymbol{\alpha}_{\text{ref}}$, and the increment-base term $\boldsymbol{\tau}_{\text{current}}$ (Eq. (21) below), so that all three terms entering the final torque command are attenuated identically at the resonant band.

A.5 Incremental torque law and RPM-based increment base

$$\delta \boldsymbol{\tau} = J (\boldsymbol{\alpha}_{\text{ref}} - \boldsymbol{\alpha}_{\text{meas}}) \quad (19)$$

$$\boldsymbol{\tau}_{\text{new}} = \boldsymbol{\tau}_{\text{current}} + \delta \boldsymbol{\tau} \quad (20)$$

$\boldsymbol{\tau}_{\text{current}}$ is estimated from measured per-motor RPM rather than the previous commanded torque:

$$F_i = k_{T,i} \cdot \text{RPM}_i^2, \quad i = 1, \dots, 4 \quad (21)$$

$$\tau_x = L (F_3 + F_4 - F_1 - F_2) \quad (22)$$

$$\tau_y = L (F_2 + F_3 - F_1 - F_4) \quad (23)$$

$$\tau_z = \kappa (F_2 + F_4 - F_1 - F_3) \quad (24)$$

where L is the motor arm length, κ is the drag-to-thrust torque ratio (motor/propeller dependent, bench/manufacturer-identified per platform), and $k_{T,i}$ is the per-motor bench-identified thrust constant. This is the inverse of the standard quadrotor force/torque allocation map: rather than assuming a model of how commanded torque produces motor speeds, the actually-realized motor speeds are read back and converted directly to the torque currently being applied — the mechanism that makes the INDI increment $\delta \boldsymbol{\tau}$ correct for whatever gap exists between the model and reality on every control cycle.

A.6 Position INDI outer loop

Active when `ctrl_mode` bit 0 is set (all three locked configurations, Appendix B). The modeled and measured net world-frame acceleration are compared directly:

$$\mathbf{a}_{\text{model}} = \mathbf{z}_{b,w} \frac{F_{\text{tot}}}{m} - g \hat{\mathbf{z}}, \quad F_{\text{tot}} = \sum_{i=1}^4 k_{T,i} \text{RPM}_i^2 \quad (25)$$

$$\mathbf{a}_{\text{meas}} = R (\mathbf{a}_{\text{body}} g) - g \hat{\mathbf{z}} \quad (26)$$

$$\mathbf{a}_{\text{indi}} = \mathbf{a}_{\text{meas}} - \mathbf{a}_{\text{model}} \quad (27)$$

where $\mathbf{z}_{b,w}$ is the body z -axis in world coordinates, $\mathbf{a}_{\text{body}}$ is the raw accelerometer reading (in g), and F_{tot} reuses the same per-motor RPM and thrust constants $k_{T,i}$ as the attitude-loop

increment base above. The desired specific force is then

$$\mathbf{f}_d = \mathbf{a}_d + K_P \mathbf{e}_p + K_V \mathbf{e}_v + K_I \int \mathbf{e}_p dt + g \hat{\mathbf{z}} + \mathbf{a}_{\text{indi}}$$

with $\mathbf{e}_p = \mathbf{p}_d - \mathbf{p}$, $\mathbf{e}_v = \mathbf{v}_d - \mathbf{v}$, and $\mathbf{a}_d$ the flatness-derived acceleration feedforward (Section 3.1). Structurally identical to the attitude increment (Eq. (19)): the model only has to be right about the mismatch between commanded and realized thrust, not the full translational dynamics.

A.7 Differential-flatness torque feedforward (planning side)

For completeness, the planning-side feedforward torque used to derive a dynamically consistent reference trajectory (Section 3.1) is the standard rigid-body inverse dynamics equation with diagonal inertia:

$$\boldsymbol{\tau}_{\text{ff}} = J\dot{\boldsymbol{\omega}} - (J\boldsymbol{\omega}) \times \boldsymbol{\omega}$$

using the flatness-derived $\boldsymbol{\omega}$, $\dot{\boldsymbol{\omega}}$ from Section 3.1's Eqs. (4)–(5). This is distinct from, and only loosely coupled to, the INDI increment law above: the planning-side torque is a purely model-based reference used to check trajectory feasibility (peak torque/thrust against platform limits), while the control-side torque command (Eq. (19)) is what is actually sent to the motor mixer.

B Full Platform Configurations

Platform	Motor type	AUW mass	Inertia $J_{xx} = J_{yy}$	Notes
Standard CF2.1	Brushed (0716)	36.4 g	$1.657 \times 10^{-5} \text{ kg m}^2$	Reference platform; lightest, lowest peak thrust
Thrust- upgraded CF2.1	Brushed (upgrade kit)	38.6 g	$\approx 1.657 \times 10^{-5} \text{ kg m}^2$	Higher peak thrust (0.18 N/motor vs. 0.13 N), heavier rotor $\rightarrow$ slower motor response
Brushless CF21BL	Brushless	41.0 g	diag(2.40, 2.40, 3.24) $\times 10^{-5} \text{ kg m}^2$	Highest motor bandwidth, DSHOT bidirectional telemetry

Table 2: Platform physical parameters. Brushless inertia from Busetto et al. [6].

Attitude (indi_gains)		Position (pos_gains)	
ctrl_mode	3 (full INDI)	kp_xy	40.0
kr, kr_z	1050.0	kp_z	30.0
kw, kw_z	87.0	kv_xy	3.3
fc_bw	70.0 Hz	kv_z	7.0
mass	0.0364 kg		
kt1..kt4	$1.4811, 1.4815, 1.5319, 1.6119 \times 10^{-10} \text{ N/RPM}^2$		

Table 3: Standard CF2.1: final locked configuration.

Attitude (indi_gains)		Position (pos_gains)	
ctrl_mode	3 (full INDI)	kp_xy	40.0
kr, kr_z	603.0	kp_z	30.0
kw, kw_z	90.0	kv_xy	8.0
fc_bw	70.0 Hz	kv_z	10.0
mass	0.0386 kg		
kt1..kt4	$2.2959, 2.4668, 2.0250, 2.5288 \times 10^{-10} \text{ N/RPM}^2$		

Table 4: Thrust-upgraded CF2.1: final locked configuration.

Attitude (indi_gains)		Position (pos_gains)	
ctrl_mode	3 (full INDI)	kp_xy	64.0
kr, kr_z	2400.0	kp_z	48.0
kw, kw_z	170.0	kv_xy	5.0
fc_bw	70.0 Hz	kv_z	7.0
filt_tau	1 (phase-matched increment base)		
mass	0.041 kg		
kt1..kt4	$4.1623, 4.0592, 4.1116, 4.0631 \times 10^{-10} \text{ N/RPM}^2$		

Table 5: Brushless CF21BL: final locked configuration.

C Extended Flight-Data Figures

The figures below are each dense multi-subplot composites (or, for the two bar charts, simply too wide to sit comfortably in the main text alongside the numeric tables) generated by the project's flight analysis tooling; they are reproduced here at full page width rather than cropped into the main text.

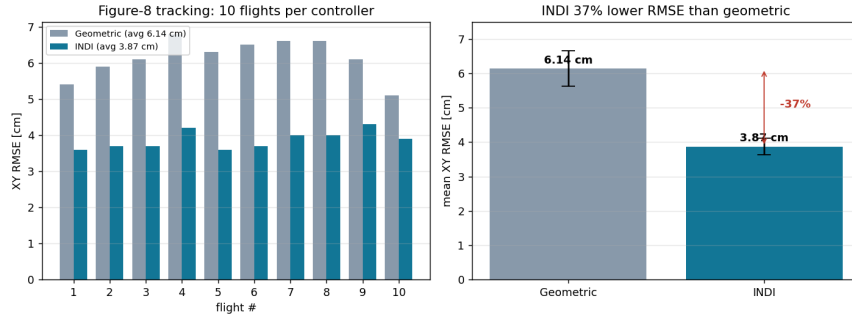


Figure 1: Standard CF2.1: geometric vs. full-INDI, 10 flights each, figure-8 at $k_t = 0.05$.

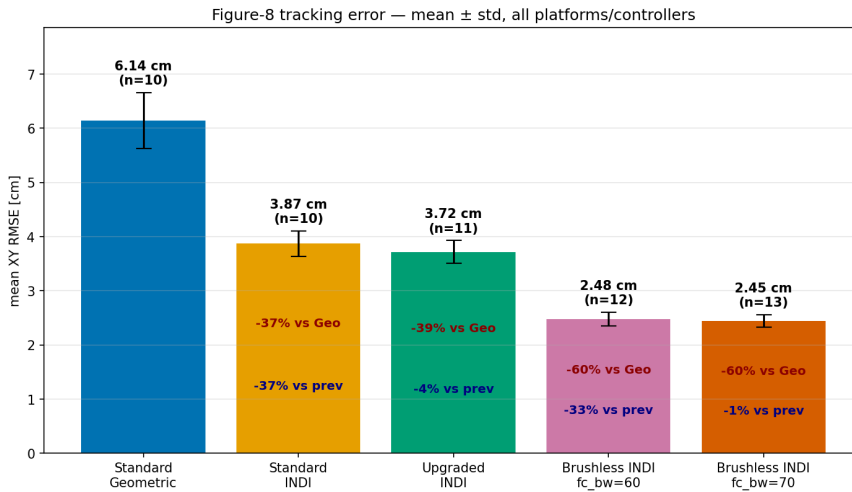


Figure 2: Five-way headline comparison: mean XY RMSE across all platform/controller configurations, figure-8 at $k_t = 0.05$.

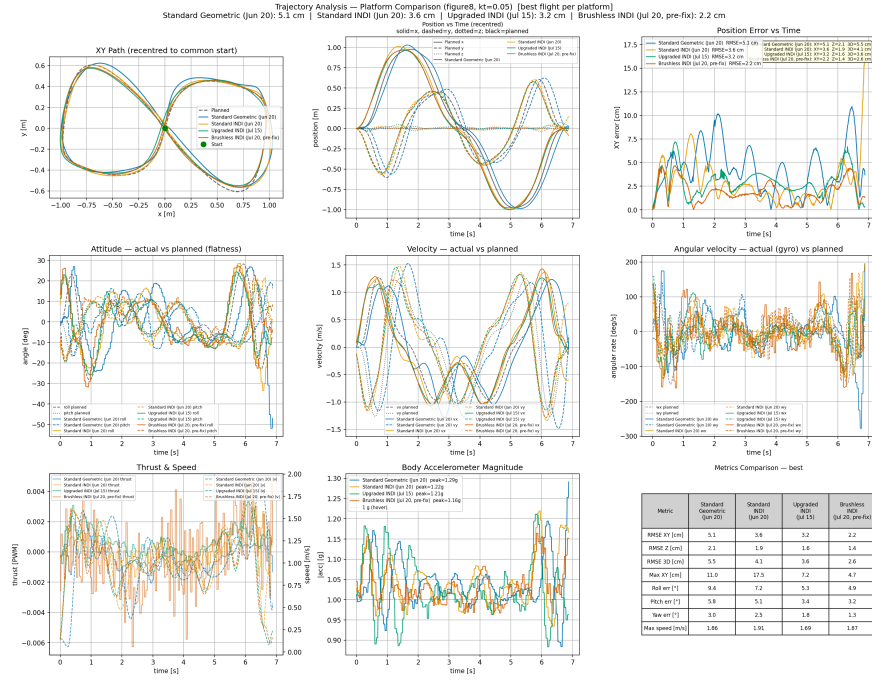


Figure 3: Best flight, brushless platform, figure-8 at $k_t = 0.05$: XY tracking overview.

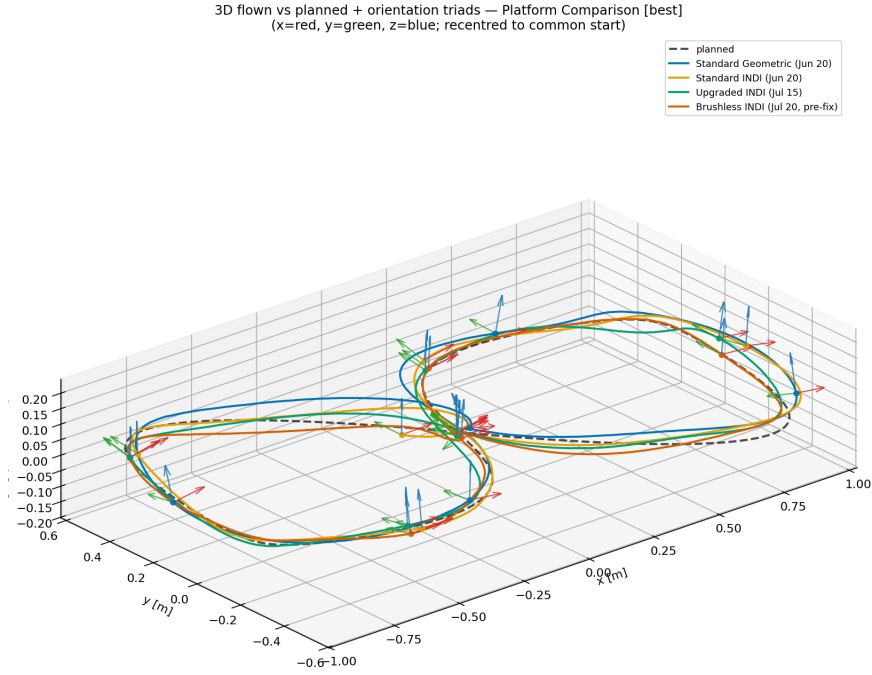


Figure 4: Best flight, brushless platform: 3D orientation.

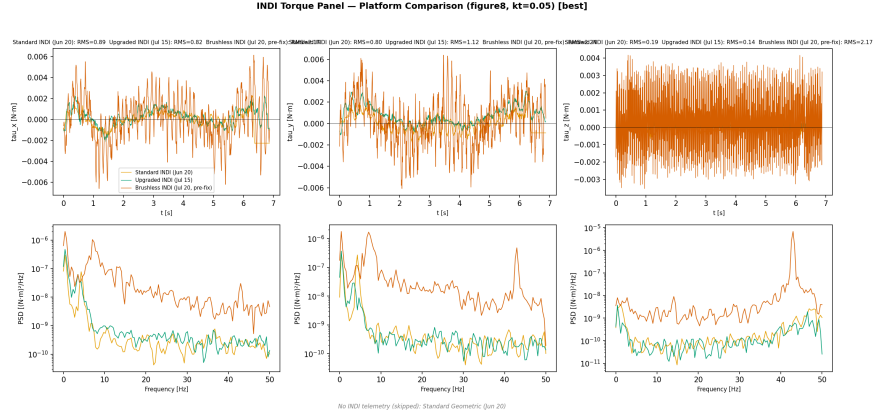


Figure 5: Best flight, brushless platform: INDI torque/angular-acceleration panel.

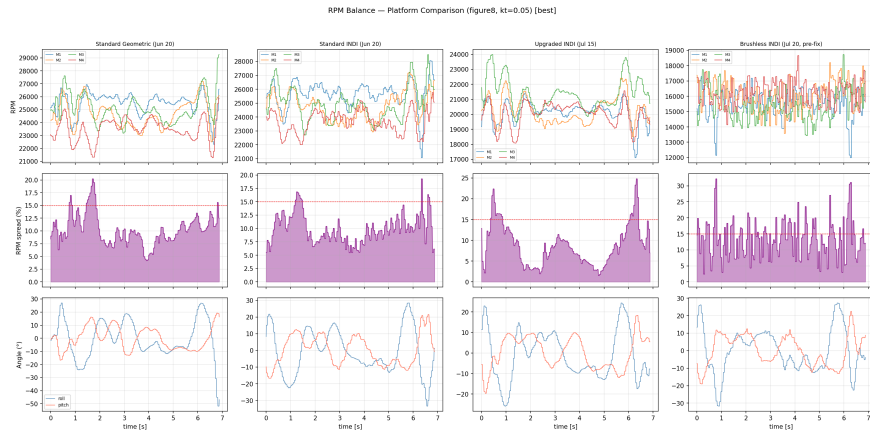


Figure 6: Best flight, brushless platform: per-motor RPM balance.

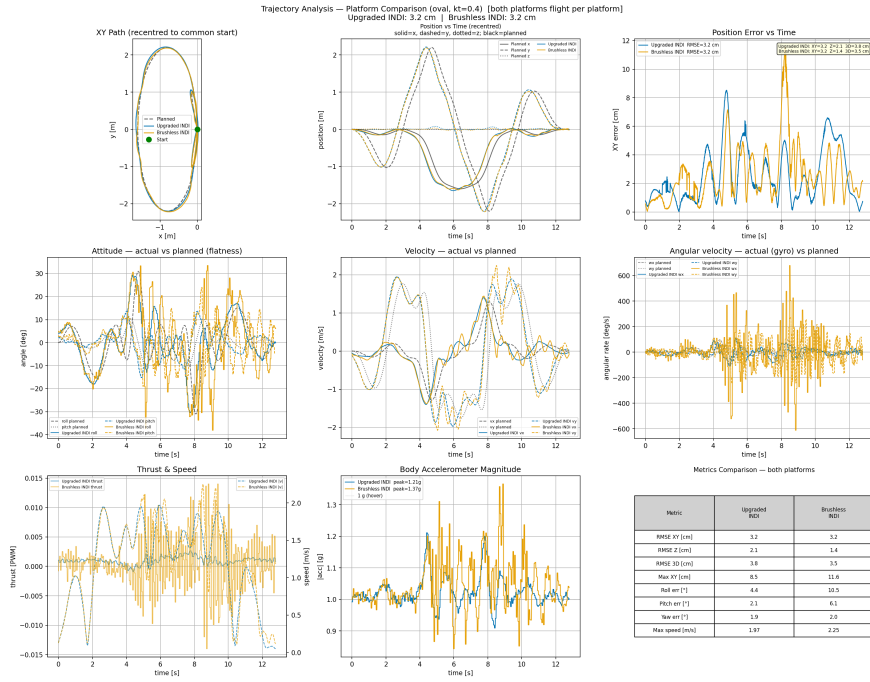


Figure 7: Upgraded vs. brushless tracking overview, oval trajectory at $k_t = 0.4$.

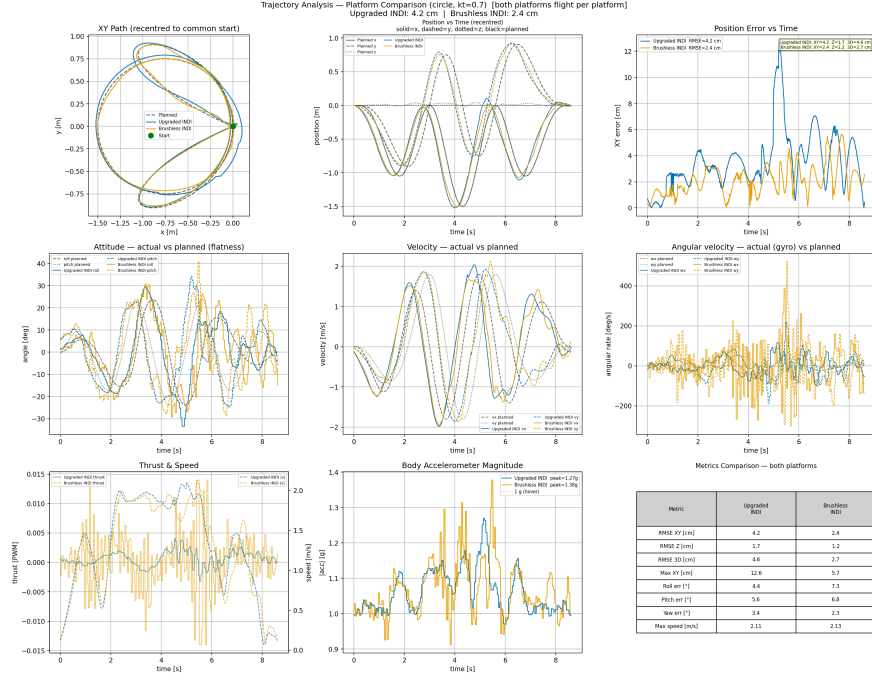


Figure 8: Upgraded vs. brushless tracking overview, circle trajectory at $k_t = 0.7$.

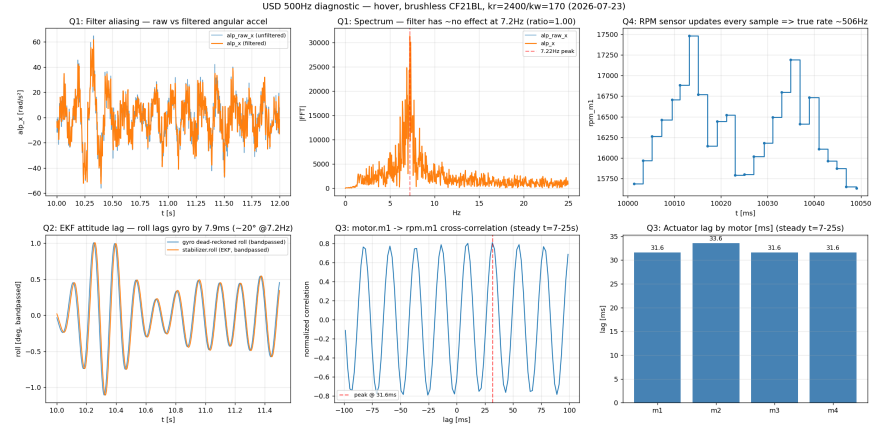


Figure 9: 500 Hz SD-card diagnostic at hover, brushless platform: the ≈ 7.22 Hz resonance quantified.

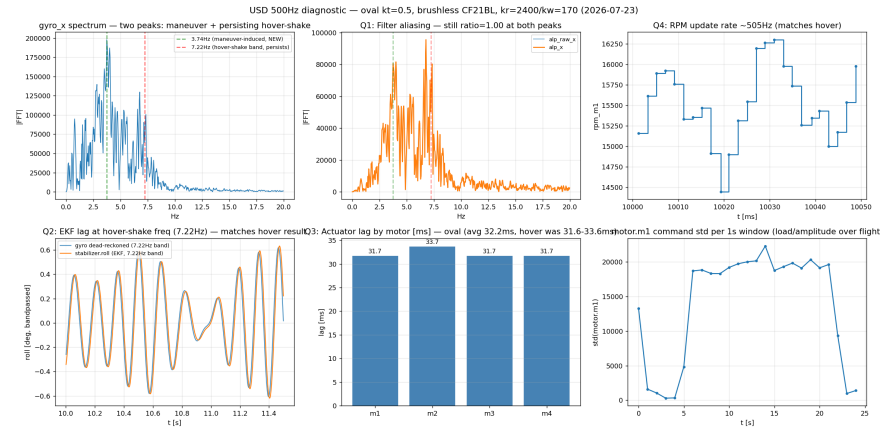


Figure 10: 500 Hz SD-card diagnostic under an oval maneuver, brushless platform: the resonance persists unchanged under real maneuvering load.

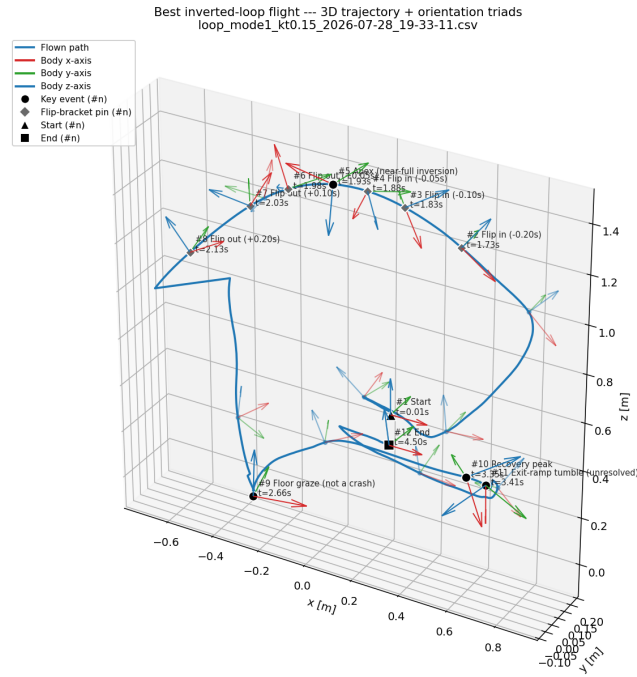


Figure 11: Best inverted-loop flight (brushless), 3D trajectory with orientation triads. Pre-flip behavior works as expected: a repeatable near-full inversion at the apex followed by a genuine recovery (arrested fall, positive climb back). Post-flip, the drone tumbles again once it reaches the trajectory’s auto-generated exit ramp – the recovery from that second tumble is the unresolved part.

D Full k_t -Sweep Data

Full per- k_t data underlying the oval and circle summaries in Section 5.3, both platforms. “Crash” entries are excluded from the RMSE/error/speed columns for that k_t .

k_t	XY / Z RMSE		Roll/Pitch/Yaw err		Max speed [m/s]	
	Upgraded	Brushless	Upgraded	Brushless	Upgraded	Brushless
0.05	2.1/0.8 cm	2.3/0.5 cm	1.7/1.3/0.8°	2.2/2.9/0.9°	1.10	1.22
0.1	2.8/0.9 cm	1.3/0.7 cm	2.3/1.5/0.7°	3.1/1.6/0.3°	1.33	1.32
0.2	2.9/1.3 cm	2.5/1.1 cm	3.3/1.5/1.0°	2.6/1.3/0.4°	1.61	1.51
0.3	3.0/1.8 cm	2.8/1.0 cm	3.2/1.7/1.4°	2.9/1.5/0.5°	1.76	1.72
0.4	3.2/2.1 cm	3.2/1.4 cm	4.4/2.1/1.9°	10.5/6.1/2.0°	1.97	2.25
0.5	crash (5/5)	3.1/1.6 cm	—	5.5/2.7/1.4°	—	2.20
0.6	crash	not flown	—	—	—	—
0.7	not flown	crash	—	—	—	—

Table 6: Oval k_t sweep, both platforms.

k_t	XY / Z RMSE		Roll/Pitch/Yaw err		Max speed [m/s]	
	Upgraded	Brushless	Upgraded	Brushless	Upgraded	Brushless
0.05	1.7/0.8 cm	1.9/0.6 cm	2.1/1.8/0.8°	1.9/1.4/0.6°	1.03	1.21
0.1	1.9/1.3 cm	1.7/0.6 cm	1.9/1.8/1.1°	4.2/3.9/0.7°	1.15	1.50
0.2	2.3/1.2 cm	6.1/1.8 cm	2.2/1.8/1.4°	14.5/13.2/3.6°	1.35	1.71
0.3	2.2/1.5 cm	1.4/0.8 cm	2.3/2.1/1.9°	3.1/2.7/0.6°	1.53	1.75
0.4	2.7/1.8 cm	4.2/1.4 cm	3.5/3.6/2.5°	10.0/9.3/2.4°	1.73	1.84
0.5	2.8/1.6 cm	3.7/1.2 cm	3.5/3.5/2.6°	7.9/7.5/2.0°	1.84	1.88
0.6	3.7/2.4 cm	1.9/1.0 cm	3.1/3.3/2.7°	4.7/4.9/1.5°	1.98	1.84
0.7	4.2/1.7 cm	2.4/1.2 cm	4.4/5.6/3.4°	7.3/6.8/2.3°	2.11	2.13
0.8	3.3/2.5 cm	crash	4.8/5.3/3.6°	—	2.37	—
0.9	crash	crash	—	—	—	—
1.0	3.0/3.6 cm	not flown	4.9/4.0/4.6°	—	2.48	—

Table 7: Circle k_t sweep, both platforms.